DSA5205 Project 1: Machine Learning in Finance

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Abstract

This report studies model complexity, regularization, and out-of-sample performance in return prediction. Task 1 reproduces the misspecified ridge experiment of Kelly, Malamud, and Zhou (2024); the remaining sections document the extensions, hidden-data predictions, and financial interpretation of neural-network generalization.

1 Task 1: Misspecified Simulation with Ridge

1.1 Design and implementation

Following Kelly, Malamud, and Zhou [1], returns are generated as

$$R_{t+1} = S_t'\beta^* + \varepsilon_{t+1}, \qquad S_t \sim \mathcal{N}(0, I_P), \qquad \varepsilon_{t+1} \sim \mathcal{N}(0,1). \tag{1}$$

The vector β^* is drawn from an isotropic Gaussian distribution and normalized so that $\|\beta^*\|_2^2 = b^* = 0.2$. The data-generating complexity is fixed at $c = P/T_{\text{tr}} = 10$. In each replication, one random permutation of the P columns is held fixed across the full grid. For a given revealed fraction q, only the first $P_1 = qP$ permuted columns are observed, so observed complexity is $cq = P_1/T_{\text{tr}}$. This nested observation rule creates misspecification while preserving the sufficiently mixed setting.

The simulation parameters are listed in Table 1 and configured in `task1_2/task1.yaml`. Data generation is implemented by `generate_data()` in `task1_2/generate_data.py`; the experiment loop and nested feature selection are in `task1_2/task1.py`.

Table 1: Task 1 simulation parameters.

Parameter	Value
Base seed and replications	5205; 50
Training and test sizes	$T_{\text{tr}} = 500$; $T_{\text{te}} = 5000$
DGP complexity and dimension	$c = 10$; $P = 5000$
Signal strength and noise	$b^* = 0.2$; $\sigma_\varepsilon = 1$
cq grid	$\{0.50, 0.75, 0.90, 0.95, 0.98, 1.02, 1.05, 1.10, 1.25, 1.50, 2, 3, 5, 7, 10\}$
Ridge grid	$z \in \{0, 0.01, 0.1, 1, 10, 50\}$

For each observed block, ridge is estimated without an intercept:

$$\widehat{\beta}(z) = \left(X'X + zT_{\text{tr}}I\right)^{-1} X'y. \tag{2}$$

At $z = 0$, the implementation uses the Moore–Penrose pseudoinverse. For $z > 0$, it solves the primal system when $P_1 \le T_{\text{tr}}$ and the equivalent dual system otherwise; see `fit_ridge()` in `task1_2/fit_ridge.py`.

All quantities are evaluated on the independent test set. With $\widehat{R}_{t+1} = S_t'\widehat{\beta}$ and timing return $R^{\pi}_{t+1} = \widehat{R}_{t+1}R_{t+1}$, the reported metrics are

$$R^2_{\text{paper}} = 1 - \frac{\mathbb{E}[(R_{t+1} - \widehat{R}_{t+1})^2]}{\mathbb{E}[R^2_{t+1}]}, \tag{3}$$

$$\mathbb{E}[R^{\pi}_{t+1}] = \mathbb{E}[\widehat{R}_{t+1}R_{t+1}], \tag{4}$$

$$\text{SR} = \frac{\mathbb{E}[R^{\pi}_{t+1}]}{\sqrt{\mathbb{E}[(R^{\pi}_{t+1})^2]}}, \qquad \|\widehat{\beta}(z)\|_2. \tag{5}$$

Their sample analogues are computed by `calculate_metrics()` in `task1_2/test_metrics.py`.

1.2 Results

Figure 1 reports averages over the 50 replications. It is the finite-sample analogue of Figures 4–6 in Kelly, Malamud, and Zhou [1], which plot population limits rather than simulated estimates. The ridgeless specification displays the expected instability around the interpolation threshold $cq = 1$: R^2_{paper} falls to about -50 at $cq = 0.98$–1.02, while $\|\widehat{\beta}\|_2$ rises above 7.5. These extremes, and timing returns above 0.02, are clipped by the displayed axes. As in the paper's Figure 4, both effects abate for $cq > 1$ because overparameterized minimum-norm regression supplies implicit regularization. Explicit shrinkage smooths the interpolation singularity; $z = 10$ or 50 keeps R^2_{paper} close to zero and the coefficient norm small.

The expected-return panel has the same broad message as the left panel of the paper's Figure 5: very simple models have low timing returns because they observe too little of the DGP, and additional features improve approximation. In the population benchmark, ridgeless expected return rises to $b^*/c = 0.02$ at $cq = 1$ and is exactly flat thereafter. The simulation approaches roughly 0.018–0.019 away from interpolation, but overshoots and fluctuates near $cq = 1$; for example, its ridgeless mean is 0.0347 at $cq = 1.02$. This difference is consistent with finite training and test samples and the nonlinear effect of near-singular estimates, rather than a contradiction of the limiting result. The paper's Figure 5 also reports portfolio volatility; the project instead requires the Sharpe ratio and coefficient norm, which expose the same interpolation instability through risk adjustment and parameter dilation.

The Sharpe panel closely reproduces the paper's Figure 6. Ridgeless and lightly regularized curves show "double ascent": Sharpe rises at low complexity, dips near interpolation as coefficient and portfolio risk explode, and rises again for $cq > 1$. Nontrivial shrinkage removes most of this dip, producing an approximately permanent ascent. At $cq = 10$, all curves converge to about 0.049–0.050; the largest value on the simulated grid is 0.0505 for $z = 50$. For $\Psi = I$, the paper's optimal shrinkage is

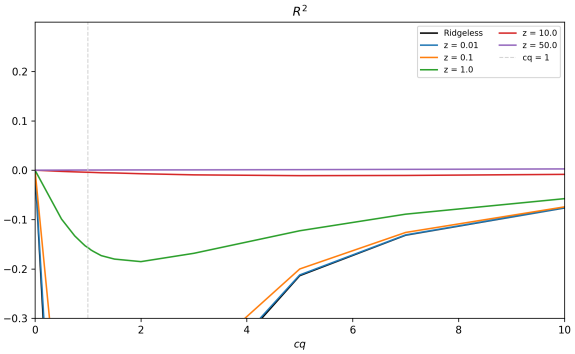
$$z^*(q) = \frac{c\{1 + b^*(1-q)\}}{b^*} = 60 - 10q, \tag{6}$$

so $z = 50$ is exactly optimal at $q = 1$ and is close to the theoretical range over the grid. Theorem 1's monotone, concave improvement applies to this optimally adjusted $z^*(q)$, not to every fixed-z curve. Overall, the finite-sample experiment supports the virtue-of-complexity mechanism: under misspecification and sufficient mixing, observing more signals improves risk-adjusted timing performance even when conventional out-of-sample R^2 remains negative.

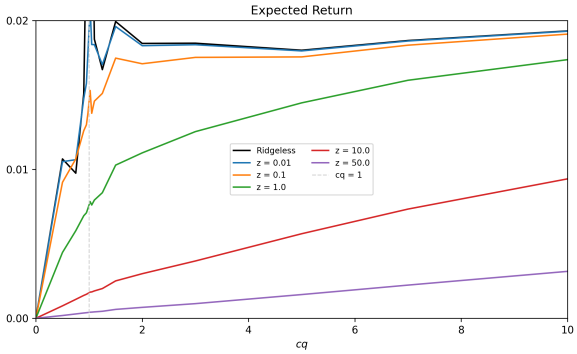
2 Task 2: Alternative Model

2.1 Method and experimental control

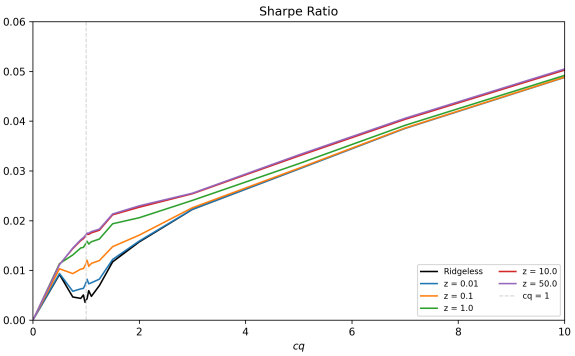
Task 2 is implemented in `task1_2/task2.py` and reads the shared configuration in `task1_2/task2.yaml`. For every replication it reuses the Task 1 seed, data-generating process, train/test



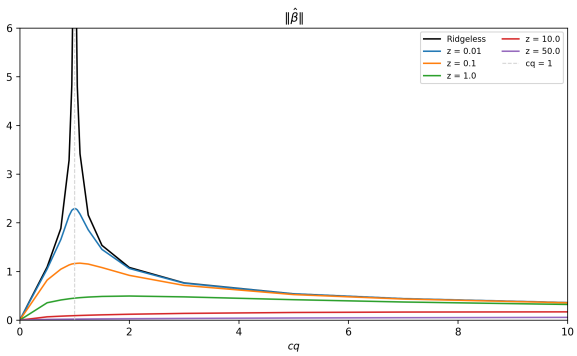
(a) R^2_{paper}.



(b) Expected timing return.



(c) Paper Sharpe ratio.



(d) Coefficient norm.

Figure 1: Finite-sample counterparts of Figures 4–6 in Kelly, Malamud, and Zhou [1], averaged over 50 replications. The dashed line marks $cq = 1$; displayed axes clip the interpolation extremes. All four panels are produced by `plot_results()` in `task1_2/plot_results.py` from `results.csv`.

split, fixed feature permutation, and cq grid, replacing ridge regression with Lasso regression. The estimator solves

$$\widehat{\beta}(\alpha) = \arg\min_{\beta} \left\{ \frac{1}{2T_{\text{tr}}} \|y - X\beta\|_2^2 + \alpha \|\beta\|_1 \right\}, \tag{7}$$

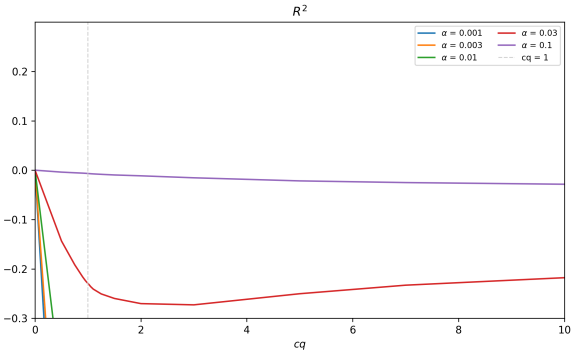
without an intercept. The fixed grid is $\alpha \in \{0.001, 0.003, 0.01, 0.03, 0.1\}$. The complete path is fitted by `lasso_path()` with warm starts, random coordinate selection, at most 3000 iterations, and tolerance 10^{-3}. The same out-of-sample metrics are then computed by `calculate_metrics()` in `task1_2/test_metrics.py`. Thus differences from Task 1 reflect the estimator rather than different simulated samples.

2.2 Results and comparison with ridge

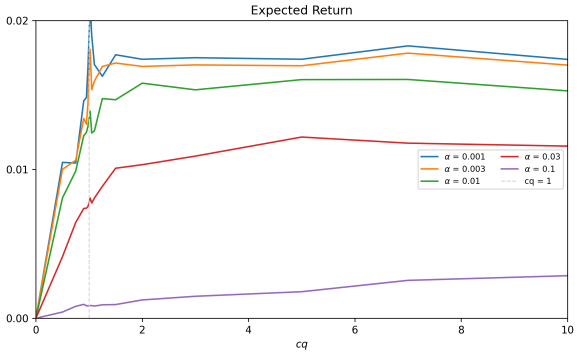
Figure 2 shows the averages over 50 replications. Weak penalties ($\alpha = 0.001$ and 0.003) retain an interpolation effect: near $cq = 1$, coefficient norms peak at approximately 2.54 and 1.71, and R^2_{paper} reaches about -5.36 and -2.41, respectively. This is substantially less extreme than ridgeless regression in Task 1, whose norm exceeds 7.5 and whose R^2_{paper} falls below -50, although it does not eliminate instability. Increasing α shrinks the norm and smooths the threshold. At $\alpha = 0.1$, R^2_{paper} remains near zero, but both expected return and Sharpe are small because the model is strongly underfit.

The economic metrics expose the bias–variance trade-off. The smallest penalty attains the largest expected timing return, 0.0211 at $cq = 1.02$, but its simultaneous prediction error keeps Sharpe at only 0.0075 there. For $cq > 1$, the weakly penalized Sharpe curves recover and reach about 0.0224 at $cq = 10$. Moderate $\alpha = 0.03$ gives the best Sharpe around the interpolation region and a relatively smooth curve, but it plateaus near 0.0197. Strong $\alpha = 0.1$ suppresses both signal and risk and reaches only 0.0129 at $cq = 10$.

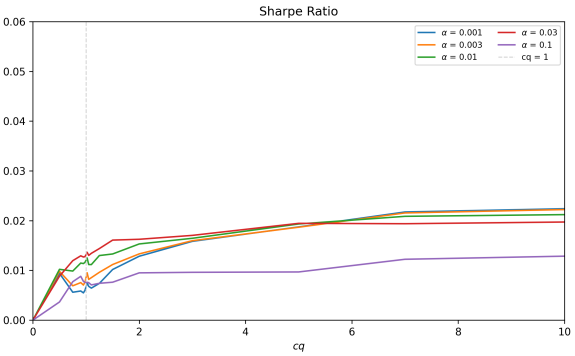
Lasso therefore reproduces the broad virtue-of-complexity pattern—timing performance generally improves as more signals are revealed—but it underperforms the best ridge specification on this dense Gaussian DGP. At $cq = 10$, the best Lasso Sharpe on the grid is 0.0224, compared with 0.0505 for ridge with $z = 50$; its corresponding R^2_{paper} is -0.390, compared with 0.0026 for ridge. This gap is consistent with model mismatch: the ℓ_1 penalty is designed to exploit sparsity, whereas the true β^* spreads a small signal across all P coordinates. Lasso discards many individually weak but collectively useful predictors, while the ℓ_2 penalty retains and stabilizes them.



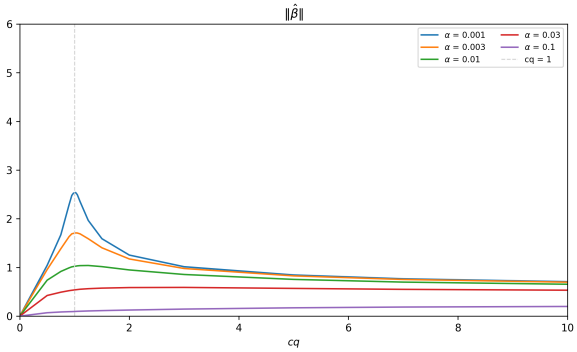
(a) R^2_{paper}.



(b) Expected timing return.



(c) Paper Sharpe ratio.



(d) Coefficient norm.

Figure 2: Out-of-sample Lasso results, averaged over 50 replications. The dashed line marks $cq = 1$ and the displayed axes clip some interpolation extremes. The panels are produced by `plot_lasso_results()` in `task1_2/plot_results.py` from `task1_2/task2_result_20261008_205656/results.csv`.

3 Task 3: Predictions for Data Sets A, B, and C

3.1 Validation and model selection

For each data set, the time index is excluded from the predictor matrix and the last 20% of the labeled observations are retained as a chronological validation block. This gives 120 validation observations for A, 48 for B, and 72 for C. Ridge, Lasso, and a one-hidden-layer neural network are fitted on the preceding observations and evaluated using the same four criteria as in Tasks 1 and 2. The primary selection criterion is fixed in advance as R^2_{paper}; expected timing return and the two Sharpe measures are retained as economic diagnostics. Fixing one selection rule is important because choosing a different model or parameter for whichever metric happens to look strongest would add another layer of validation overfitting, especially for B and C, whose holdout blocks are small.

The initial grids identify the relevant region for each estimator. The selected model is then exploited in three successively narrower searches, reported in Table 2. Each round centers the next grid on the preceding maximizer of R^2_{paper}. The search is stopped after the third round because the remaining differences are small relative to the size of the validation blocks. After selection, the chosen specification is refitted on all labeled observations and used to produce the hidden-data predictions.

Table 2: Task 3 hyperparameter exploitation rounds.

Data set	Model	Round	Search grid	Best parameter
A	Ridge	1	$z \in [1.5, 5.0]$	4.0
A	Ridge	2	$z \in [3.4, 4.6]$, step 0.2	3.8
A	Ridge	3	$z \in [3.65, 3.95]$, step 0.05	3.75
B	Lasso	1	$\alpha \in [0.10, 0.14]$, step 0.005	0.120
B	Lasso	2	$\alpha \in [0.116, 0.124]$, step 0.002	0.122
B	Lasso	3	$\alpha \in [0.1200, 0.1240]$, step 0.0005	0.1215
C	Lasso	1	$\alpha \in [0.15, 0.21]$, step 0.01	0.1900
C	Lasso	2	$\alpha \in [0.1800, 0.2000]$, step 0.0025	0.1875
C	Lasso	3	$\alpha \in [0.1850, 0.1900]$, step 0.0005	0.1865

Data set A. Data set A has 600 observations and 600 predictors. Ridge is the only candidate model with positive validation R^2_{paper}. The final local search has a broad, nearly flat maximum over $z = 3.7$–3.9, with the largest value, 0.042628, at $z = 3.75$. This flatness supports selecting the interior value rather than extending the search to an unnecessarily precise decimal. The corresponding timing return and paper Sharpe are 0.056045 and 0.171803. In comparison, the best Lasso specification in the candidate experiment has $R^2_{\text{paper}} = -0.010871$ and negative timing and Sharpe measures. The selected neural network has a positive timing return of 0.144969 but $R^2_{\text{paper}} = -0.105352$, indicating that its output scale is poorly calibrated despite some directional information. Ridge therefore gives the best balance of prediction accuracy and economically positive timing performance for A.

Data set B. Data set B is strongly overparameterized, with 240 observations and 2400 predictors. Lasso is preferred because its coordinate-wise shrinkage can discard irrelevant predictors rather than retaining all 2400 directions. The three exploitation rounds move the local maximizer from 0.120 to 0.122 and finally to $\alpha = 0.1215$. At this value, $R^2_{\text{paper}} = 0.001634$, expected timing return is 0.059076, and the paper Sharpe is 0.199729. The nearby value $\alpha = 0.120$ has marginally higher timing and Sharpe measures but marginally lower R^2_{paper}; 0.1215 is retained to preserve the pre-specified selection rule. Ridge produces $R^2_{\text{paper}} = -0.061635$ together with negative timing and Sharpe measures, while the selected neural network gives $R^2_{\text{paper}} = -0.465629$. Thus the choice of Lasso is not based only on a fine difference inside its local grid: it also dominates the alternative model classes on validation prediction accuracy and both economic metrics.

Data set C. Data set C contains 360 observations and 1800 predictors, so sparsity is again more plausible than a dense unrestricted fit. The first Lasso experiment used $\alpha = 0.15$ and produced $R^2_{\text{paper}} = -0.001020$; a much stronger penalty, $\alpha = 0.30$, eliminated every coefficient and reduced the model to the zero-prediction benchmark. Expanding and then narrowing the non-degenerate region yields $\alpha = 0.1865$, where $R^2_{\text{paper}} = 0.004597$ and all three economic metrics are positive. By contrast, the best Ridge candidate has $R^2_{\text{paper}} = -0.073618$ and negative timing performance, and the selected neural network has $R^2_{\text{paper}} = -0.724517$. The final Lasso penalty therefore represents the interior compromise between the noisy, weakly penalized solutions and the all-zero solution generated by excessive shrinkage.

3.2 Final models and prediction files

Table 3 records the specifications selected under the common validation rule. The reported prediction files are generated only after refitting these specifications on the full labeled samples; their row counts agree with the corresponding hidden feature files, as shown in Table 4.

Table 3: Task 3 final validation results selected by R^2_{paper}.

Data set	Model and parameter	R^2_{paper}	$\mathbb{E}[R^\pi]$	SR_{paper}	$\text{SR}_{\text{variance}}$	$\|\widehat{\beta}\|_2$
A	Ridge, $z = 3.75$	0.042628	0.056045	0.171803	0.174396	0.233292
B	Lasso, $\alpha = 0.1215$	0.001634	0.059076	0.199729	0.203836	0.320890
C	Lasso, $\alpha = 0.1865$	0.004597	0.003104	0.091762	0.092151	0.037342

Table 4: Task 3 final prediction files.

Data set	Observations	Prediction file
A	1000	`task3/A_ridge_result_20261010_155038/predictions_A.csv`
B	4000	`task3/B_lasso_result_20261010_155133/predictions_B.csv`
C	2000	`task3/C_lasso_result_20261010_155221/predictions_C.csv`

4 Task 4: Generalization and Overfitting in Neural Networks

4.1 Main finding of Montanari and Urbani

4.2 Implication for financial machine learning

5 Task 5: Video and Reproducibility

6 Conclusion

References

[1] Kelly, B. T., Malamud, S., and Zhou, K. (2024). The virtue of complexity in return prediction. *Journal of Finance*, 79(1), 459–503.

[2] Montanari, A. and Urbani, P. (2025). Dynamical decoupling of generalization and overfitting in large two-layer networks. *Advances in Neural Information Processing Systems*, 38.